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WASHA TAKWIMU
WORKING ON APPLICATIONS FOR DATA SCIENCE AND HEALTH IN AFRICA

WASHA Takwimu (Swahili for Ignite Data)
Working on Applications for Data Science and Health in Africa

Mini project

April 2026

Mini-Project: Application of Supervised Machine Learning

- **Aim of the mini project**

To apply supervised machine learning methods to real-world data from four countries (South Africa, Ghana, Tanzania, and Uganda) to predict selected health or demographic outcomes and interpret key predictors.

Overall learning objectives

By the end of the project, participants should be able to:

1. Formulate a clear prediction question using a binary outcome from the IR file.
2. Select, clean, and recode relevant predictor variables (features) from the women of reproductive age.
3. Split data into training and test (and, if desired, validation) sets.
4. Fit at least two machine-learning models (e.g., random forest, gradient boosting).
5. Assess model performance using appropriate metrics (e.g., accuracy, sensitivity, specificity, ROC/AUC).
6. Interpret the most important predictors and discuss public-health implications and limitations.

Tasks for each group

1. Clearly define the outcome variable (Y) and show how it is coded (0/1).
2. Choose and justify a set of predictor variables (X) from the women of reproductive age; recode and handle missing values as needed.
3. Briefly explore the data (descriptive statistics and simple plots).
4. Split the dataset into training and test sets.
5. Fit at least one non-linear ML model (e.g., random forest, gradient boosting, or another agreed method).
6. Compare models on the test set using at least two performance metrics.
7. Summarize key findings, limitations, and possible policy/program implications.

Description of the dataset

We will use the unclean dataset from four countries

The file is for **women of reproductive age (usually 15–49 years)**

The file consists of:

1. Sociodemographic variables (age, education, marital status, residence, wealth, etc.)
2. Reproductive history and fertility
3. Family planning knowledge and use
4. Maternal and child health (ANC, delivery, PNC, vaccinations—often via children's histories)

Group 1-UGANDA

Group 1 – Predict modern contraceptive use

Outcome (Y):

- Current use of a modern contraceptive method (yes/no)
  - e.g. variable based on v312/v313: modern = 1, traditional/none = 0.

Key features (X):

- Socio-demographic
  - Age (v012, and age group)
  - Education level (v106 or years v133)
  - Marital status (v501)
  - Residence (urban/rural, v025)
  - Region (v024)
  - Wealth index (v190)

Key features (X):

- Reproductive history
  - Number of living children (v218)
  - Recent birth in last 5 years (yes/no, from v208 or birth history)
  - Age at first birth (v212)
  - Desire for more children (v605)
  - Ideal number of children (v613)
- Knowledge and attitudes
  - Knowledge of any modern method (v304 series or v380)
  - Exposure to FP messages in media (v384a–c, or v157–v159)
  - Decision-making power (v743a–f)

Group 1 - Members

First name	Surname
Yekonia	Mlowe
Gilbert	Waria
Sibusisiwe	Dlamini
Linda	Paulo
Shoma	Benjamini

Group 2 - TANZANIA

Group 2 – Predict four or more ANC visits for last birth

- **Outcome (Y):**
- Had ≥4 ANC visits for the most recent birth (yes/no)
  - From m14 (number of ANC visits) for last birth.

Key features (X):

- Woman's background
  - Age at last birth (using v012 and child's date)
  - Education (v106 / v133)
  - Wealth index (v190)
  - Residence (v025)
  - Region (v024)

Key features (X):

- Reproductive + obstetric history
  - Parity (v218)
  - Previous child death (any child with b5 = 0)
  - Birth interval preceding last birth (b11)
- Access and exposure
  - Distance or problem in getting care (v467b or similar)
  - Media exposure (v157–v159)
  - Partner's education (v701)

Group 2 - Members

First name	Surname
Alfateresia	Mwasangama
Antidius	Rwehumbiza
Happines	Kajoka
James	Serubugo
Liberata	Mwita

Group 3 – SOUTH AFRICA

- **Group 3 – Predict experience of intimate partner violence (IPV)**

(Use only if IPV module is available

- **Outcome (Y):**
- Any physical or sexual IPV in the last 12 months (yes/no)
  - From d105–d108 or equivalent.
- **Key features (X):**
- Socio-demographic
  - Age (v012)
  - Education (v106 / v133)
  - Wealth index (v190)
  - Residence (v025)
  - Region (v024)
  - Partner's education (v701)

Key features (X):

- Relationship factors
  - Marital status (v501)
  - Age difference with partner (v730 or constructed)
  - Partner's alcohol use (d113, if available)
- Gender norms and empowerment
  - Woman's decision-making power (v743a–f)
  - Attitudes approving wife-beating (v744a–f)
  - Attitudes to refusing sex (v632, v632a)

Group 3- Members

First name	Surname
Bruno	Mmbando
Esther	Nhang’ano
Kasim	Chambulilo
Alfred	Bortey
Innocent	Mboya

Group 4 -GHANA

Group 4 - Unmet need for family planning (yes/no)

Use the DHS variable for unmet need if present (often **v626a or v626** recoded to unmet vs no unmet).

Key predictors (X)

- Age (**v012**), marital status (**v501**)
- Education (**v106** or **v133**)
- Wealth index (**v190**), residence (**v025**), region (**v024**)
- Number of living children (**v218**)
- Fertility preferences (**v605**, **v613**)
- Media exposure (**v157**, **v158**, **v159**)
- Decision-making autonomy (**v743a–v743f** combined)

Group 4- Members

First name	Surname
Julius	Adjei-Roger
Frank	Mapend
Frank	John
Emmanuael	Deogratius
Irene	Namata
Martha	Magili

THANK YOU

Questions??